KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) What does protocol achieve? Differentiate between intranet and extranet. (10)
   b) Why do we need Web caching? Discuss about the consistency of Web caching. (10)
   c) What is P2P architecture. Describe what transport service does an app need? (10)

2. a) Compare among WWW, HTTP and Internet. Show the HTTP request and response message format. (10)
   b) Suppose, *Alice* (ID – "alice@net.edu") wants to send message to *Bob* (ID – "bob@denza.edu"). Show SMTP interactions between SMTP client and SMTP server for this communication. (20)

3. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (14)
   b) How does RSA provide digital signature mechanism? (10)
   c) Justify that – "Switch is more intelligent than hub". (06)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Why do we need Network Address Translation (NAT)? Briefly explain how does NAT work. (10)
   b) Compare the contrast link-state and distance vector routing algorithms. (09)
   c) Suppose a source and a destination are connected through a switch and a router. If the source sends a message to the destination, then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above-mentioned devices. Explain with necessary figure. (11)

5. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) What is the relation between window size and sequence number in case of selective repeat? Why this relation is needed? (09)
   c) List down the steps involved in SSL handshaking. Also present a pictorial view of packet transmission. (10)

6. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) What is Ethernet? Describe how does CSMA/CD protocol work with a detail look on exponential back-off algorithm. (12)
   c) Suppose, a company has 200 personal computers and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PCs. How personal computer get the IP address if you have a network address: 172.16.0.0/24. Also, state the dynamic range of IP addresses. (12)